William Jackson

Senior Software Engineer - Go

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PROFESSIONAL SUMMARY

Senior Software Engineer with ~10 years of experience delivering resilient backend systems, data-intensive platforms, and production AI/ML services across the **telemetry, analytics, and intelligent automation industry** using **Go 1.18–1.23, Python 3.8–3.12, and TypeScript 4.x–5.x**. Strong expertise in **event-driven architectures**, **gRPC/REST platforms**, and reliability engineering, with consistent success reducing duplicate processing, stabilizing distributed workflows, and improving incident response through **OpenTelemetry**, structured logging, and actionable metrics. Known for modernizing brittle pipelines, hardening cloud services, and implementing safe rollout, retry, and backpressure strategies that keep high-volume systems observable, scalable, and operationally predictable.

SKILLS

- **Languages & Core — Go 1.18–1.23**, **Python 3.8–3.12**, **TypeScript 4.x–5.x**, JavaScript ES6+, SQL, Bash, Linux, concurrency patterns, memory profiling, performance tuning
- **Backend & APIs — gRPC**, **Protobuf**, **REST APIs**, middleware/interceptors, retries, timeouts, circuit breaking, rate limiting, backpressure, background workers, idempotency design, pagination, API versioning, WebSockets, service profiling
- **Frontend — React 17–18**, **Next.js 12–14**, Redux, HTML5, CSS3, Tailwind CSS, Webpack, Vite, component-driven UI patterns, internal tooling interfaces
- **Data, Messaging & Search — PostgreSQL**, MySQL, SQL Server, Redis, MongoDB, DynamoDB, **AWS SQS/SNS**, **Kafka**, RabbitMQ, **Elasticsearch**, **OpenSearch**, Redshift, BigQuery, Parquet, feature pipelines, dataset lineage, analytics event modeling
- **Cloud & DevOps — AWS**: EKS, ECS, EC2, Lambda, API Gateway, S3, CloudFront, RDS, DynamoDB, ElastiCache, EventBridge, Step Functions, IAM | **Azure**: AKS, App Service, Azure SQL, Service Bus, Key Vault, Application Insights | **GCP**: GKE, Cloud Run, Pub/Sub, Cloud Storage, BigQuery | Docker, Kubernetes, Helm, Argo CD, GitOps, Terraform, CloudFormation, GitHub Actions, GitLab CI, Jenkins, Azure DevOps, CircleCI
- **Observability, Quality & Security — OpenTelemetry**, Datadog, Prometheus, Grafana, ELK/EFK, CloudWatch, structured logging, audit logging, Jest, Mocha, Chai, React Testing Library, Cypress, Playwright, API contract testing, **OAuth2**, **OpenID Connect**, **JWT**, Okta, Auth0, AWS Cognito, RBAC, ABAC, secrets handling, rate limiting

WORK HISTORY

Senior Software Engineer
February 2021 to December 2025

Viaro Networks Inc. - Delaware, United States

- Architected a high-volume **Go** ingestion platform for bursty telemetry workloads, implementing bounded worker pools, replay-safe consumption, and backpressure controls that preserved downstream stability during producer spikes and partial infrastructure outages.
- Redesigned duplicate protection logic with **idempotency keys**, dedupe windows, and offset-aware processing, cutting double-counted event scenarios by more than **90%** during retry storms and broker rebalancing events.
- Modernized a fragile legacy **Python** scoring flow into a reproducible training and inference pipeline with immutable configuration, snapshot-based dataset handling, and controlled runtime inputs, which materially improved auditability and rollback confidence.
- Built a unified **gRPC** and **REST** access layer with shared timeout handling, request validation, authentication middleware, and standardized error envelopes, significantly reducing integration ambiguity for partner-facing consumers.
- Implemented centralized **OAuth2 / OpenID Connect / JWT** enforcement and policy-driven **RBAC/ABAC** checks, resolving inconsistent authorization behavior that had previously created edge-case access bugs across internal services.

- Introduced end-to-end **OpenTelemetry** tracing with correlation IDs and latency segmentation, allowing the team to isolate cross-service bottlenecks faster and improve mean time to detect production regressions by roughly **50%**.
- Designed a contract-governed feature storage model using **PostgreSQL** and **S3**, adding ownership metadata and validation rules that prevented silent schema drift from degrading downstream model behavior.
- Delivered safe canary rollouts for inference services on **Kubernetes** using health gates, traffic sampling, and comparative error monitoring, reducing risky production releases and improving rollout confidence by more than **40%**.
- Resolved recurring dashboard slowdowns by batching high-cardinality writes, reworking storage access patterns, and tuning **PostgreSQL** indexes, which kept operational dashboards responsive even under peak traffic windows.
- Implemented resilient reconciliation workers using **AWS SQS/SNS** with dead-letter handling and replay controls, eliminating many manual rerun tasks that had been required whenever upstream systems became unstable.
- Strengthened operational readiness with structured logging, field-level redaction, **CloudWatch** alerting, and **ELK/EFK** investigation workflows, giving on-call engineers faster and safer incident triage during customer-facing failures.
- Automated build, test, packaging, and deployment paths through **GitHub Actions** and **Jenkins**, creating repeatable release mechanics, safer rollback procedures, and more predictable delivery of backend services.

Software Engineer October 2018 to January 2021
Avantia, Inc. - Ohio, United States

- Built production **Python** ETL pipelines that normalized inconsistent operational feeds into training-ready datasets, enforcing stable identifiers and timezone-safe timestamp handling to support reliable analytics and model preparation.
- Implemented deterministic preprocessing routines for missing values, category normalization, and feature encoding, reducing evaluation volatility by more than **30%** when source data distributions shifted unexpectedly.
- Created expectation-style validation layers with quarantine outputs and failure reporting, catching schema mismatches early and preventing bad records from silently contaminating downstream training and scoring workflows.
- Designed and tuned **PostgreSQL** schemas, indexes, and materialized extraction paths for nightly feature generation, resolving several SLA risks caused by heavy joins and poorly selective query patterns.
- Added incremental backfill tooling with watermark-based checkpoints and replay guards, enabling safe historical reprocessing without duplicate inserts or unnecessary stress on source APIs during recovery events.
- Developed a lightweight **Go** command-line utility that generated reproducible dataset slices for analysts, reducing manual SQL export work and improving consistency across experimentation and validation tasks.
- Instrumented internal services with correlated metrics and structured logs so a single customer’s data path could be traced across ingestion, transformation, and scoring jobs during root-cause analysis.
- Built internal scoring APIs with clear **REST** error contracts, retries, timeouts, and circuit-breaker style safeguards, preventing upstream instability from cascading into broader service disruptions.
- Improved runtime efficiency by optimizing hot paths with **NumPy**, reducing I/O waste through **Parquet**, and freeing enough compute capacity to support additional experiments without proportional infrastructure cost growth.
- Introduced a lightweight model registry process with immutable artifacts, promotion metadata, and deployment references, ensuring released models always pointed to traceable inputs and reproducible build outputs.
- Led structured post-incident reviews for pipeline failures and recurring data defects, translating root causes into guardrails and steadily reducing repeated operational issues across the analytics delivery lifecycle.

Software Developer March 2016 to September 2018
ArnAmy, Inc. - Florida, United States

- Developed **Python** ETL workflows that consolidated multi-vendor event feeds, applying strict deduplication and timezone normalization so downstream reporting reflected consistent business reality across regions.

- Built internal **REST APIs** exposing curated datasets with documented pagination and stable response contracts, preventing fragile client integrations and reducing repeated support requests around large-result retrieval.
- Added regression protection around transformation logic using golden datasets and fixture-based checks, which preserved historical comparability and caught subtle behavioral regressions before release.
- Introduced **Docker**-based local development environments to eliminate machine-specific dependency issues, making defect reproduction faster and significantly reducing onboarding friction for new engineers.
- Optimized heavy reporting queries across **MySQL** and **PostgreSQL** through index design, join rewrites, and query-plan review, shrinking batch windows and improving report freshness for internal stakeholders.
- Implemented retry-aware background jobs with idempotent processing behavior, removing many manual rerun steps that had previously been needed after transient failures or partial external timeouts.
- Added service health checks, runtime diagnostics, and pragmatic **Linux** operational routines that surfaced failures earlier and reduced the duration of user-visible disruption during recurring incidents.
- Built reconciliation utilities that compared source counts with processed outputs, surfacing drift conditions early and preventing long-running discrepancies from quietly affecting stakeholder reporting.
- Improved maintainability by extracting shared parsing and transformation rules into reusable modules, reducing duplication and making enhancements safer across multiple ingest paths.
- Prepared systems for sustained growth by partitioning high-volume tables and introducing archival routines, keeping storage expansion predictable while preserving acceptable query performance over time.

Junior Software Developer
October 2014 to February 2016
Centric Tech Inc. - New Jersey, United States

- Wrote **Python** and **SQL** utilities to clean, validate, and reconcile inconsistent customer datasets, adding defensive checks that stopped recurring export failures caused by format drift.
- Built small **Bash** automations for nightly pulls, comparisons, and delivery preparation, reducing spreadsheet-heavy manual work and creating more consistent audit-friendly outputs for the team.
- Implemented anomaly summaries and statistical checks for operational metrics, helping business users identify data quality problems before they influenced planning or customer-facing reporting.
- Standardized lightweight **Git** review practices and change checklists, improving collaboration quality and reducing avoidable merge-related defects during fast-moving delivery cycles.
- Improved scheduled job reliability through clearer exit-code handling, structured failure summaries, and practical troubleshooting guidance that made support work less guesswork-driven for engineers.
- Partnered with senior developers to document source systems, data ownership, and usage assumptions, reducing duplicate effort and clarifying definitions that had previously caused reporting confusion.
- Optimized a slow reporting workflow by redesigning table access patterns and adding appropriate indexing, making iterative analysis practical where runtime had previously blocked regular use.
- Supported safer incremental delivery by breaking larger changes into smaller deployable units, which reduced regression risk and made post-release troubleshooting materially easier.
- Contributed to production issue resolution on **Linux** hosts with pragmatic runbooks and recurring-failure notes, improving response consistency and helping the team recover faster from operational issues.

EDUCATION

Bachelor’s degree in Computer Science
2010 to 2014
Harvard University

- Focused on strong fundamentals in **software engineering**, **databases**, and **systems**, building a foundation that translates into production-first delivery and disciplined operational practices.